

Human Body Systems

Circulatory + Nervous Systems



Let's explore: how does your body work?

Your body breathes, moves, thinks, feels, eats, and responds every day. But how does it transport materials and control activities?



BREATHE



MOVE



THINK



FEEL



EAT



RESPOND

CIRCULATORY =
transports

NERVOUS = controls

The circulatory system — what it does

Transports oxygen, nutrients, and waste throughout the body.

HEART

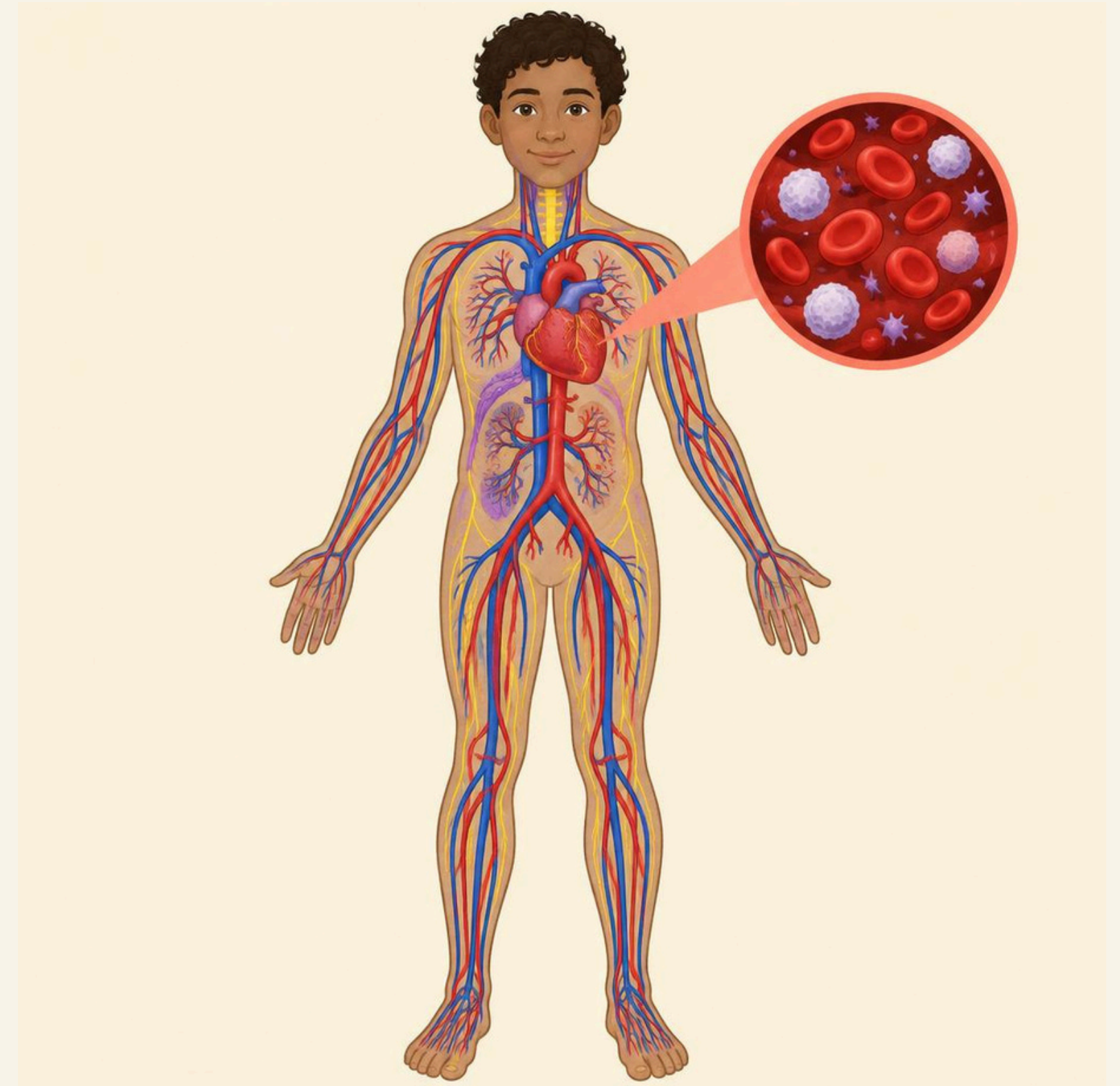
pumps blood

BLOOD

carries oxygen, nutrients + waste

BLOOD VESSELS

carry blood everywhere



Blood vessels: three pathways



ARTERIES

Carry blood **AWAY**
from the heart



VEINS

Carry blood
TOWARD the heart



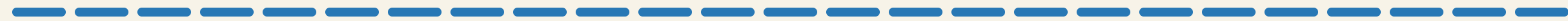
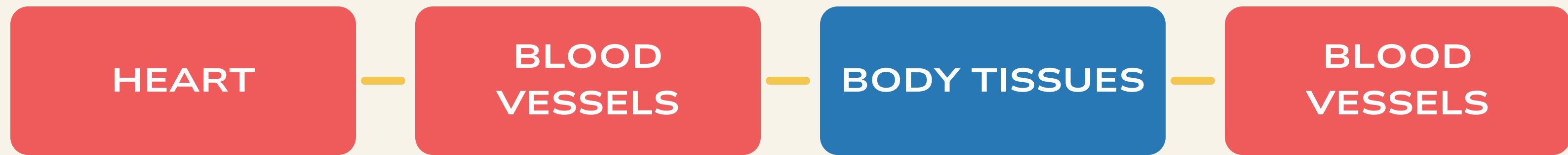
CAPILLARIES

Tiny exchange
network at body
tissues

Memory trick: A = Arteries = Away

Heart to body — and back again

DELIVER oxygen + nutrients



RETURN waste to be removed

RUNNING? Muscles need more oxygen, so your heart beats faster.

The nervous system — what it does

Controls and coordinates thinking, feeling, moving, remembering, and responding.

BRAIN

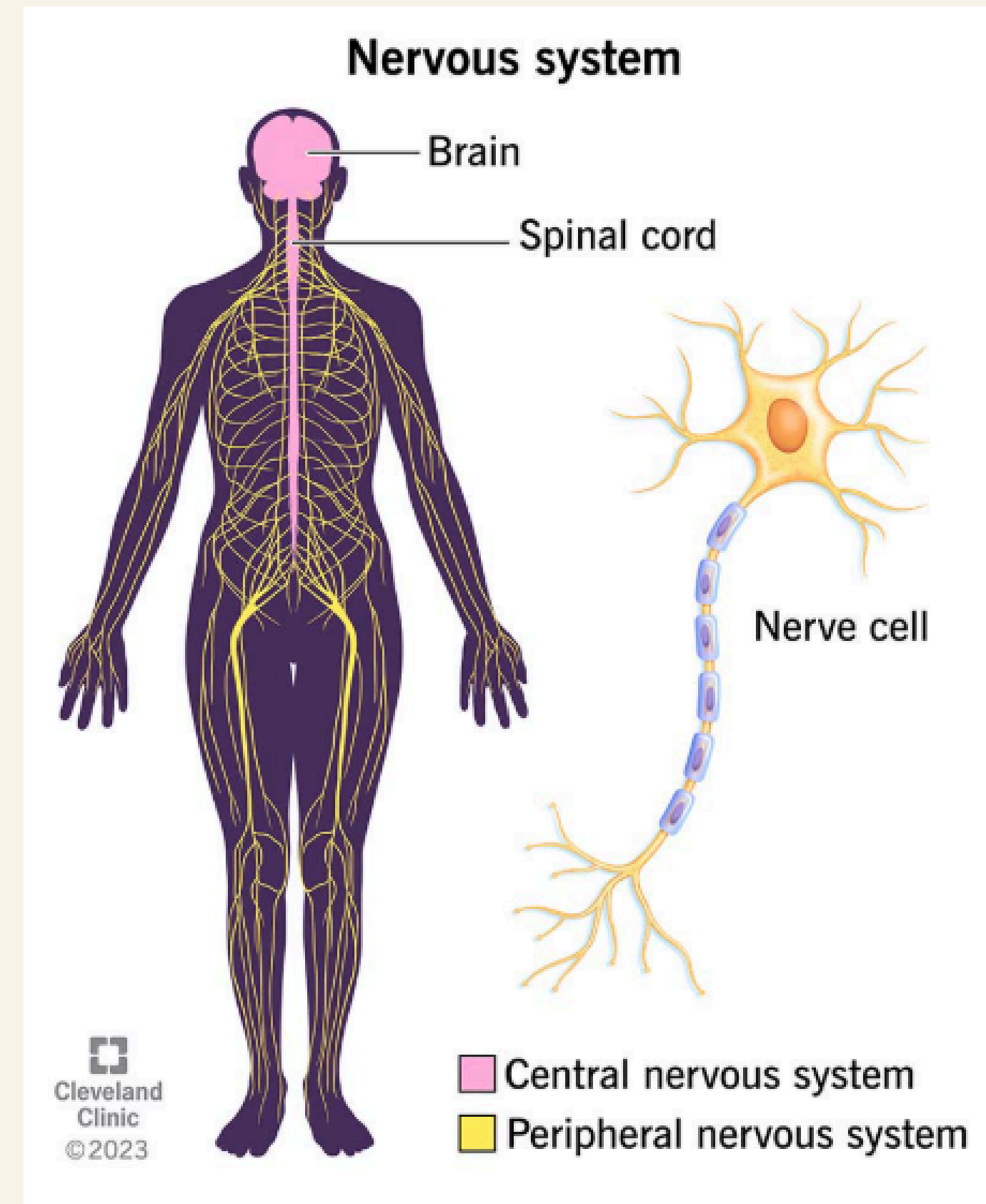
control center

SPINAL CORD

message highway

NERVES

messages to + from body parts

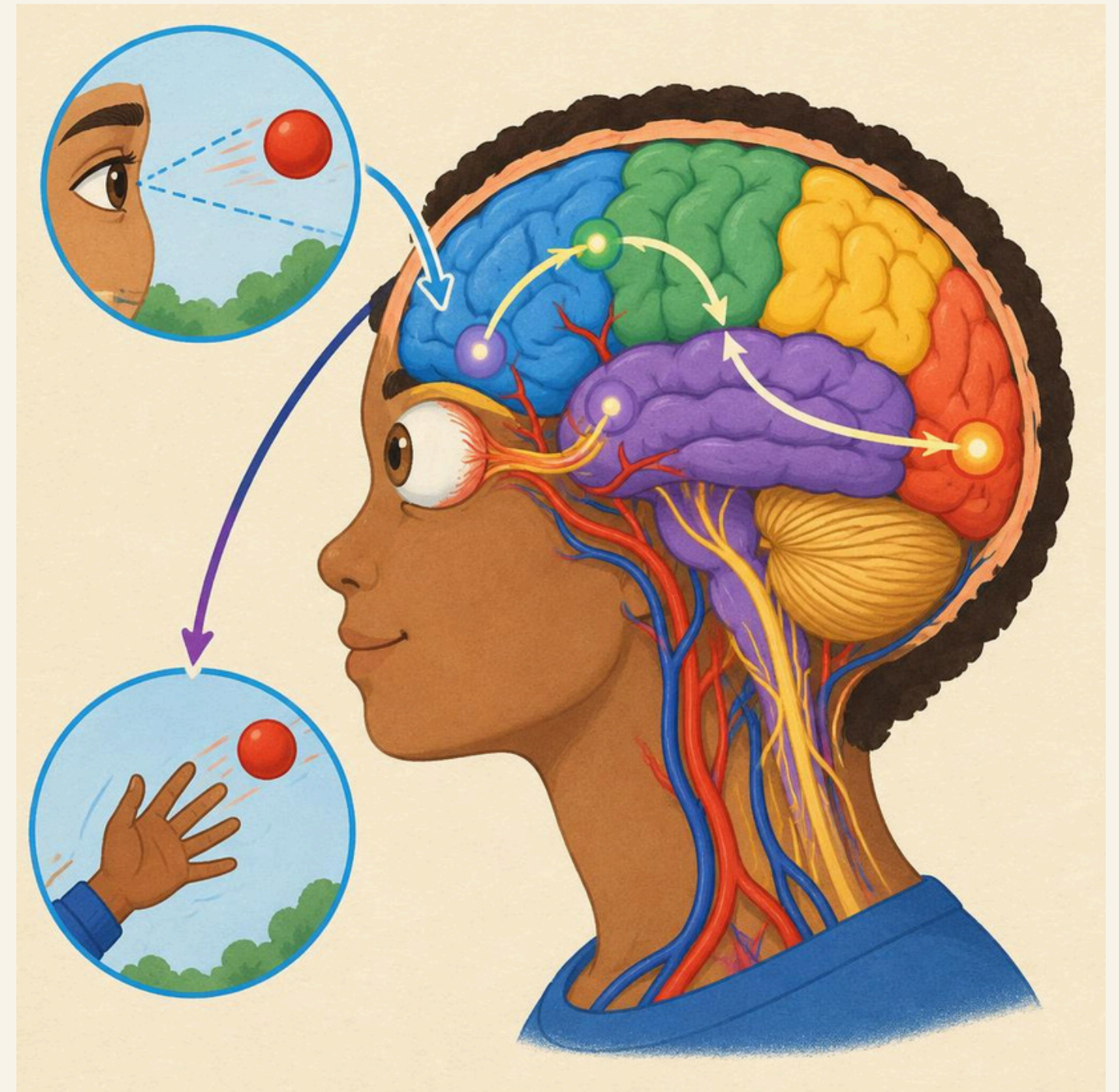


The brain turns information into action

THINK • LEARN • REMEMBER

The brain interprets your senses and controls movement.

SEE the ball PROCESS MOVE to catch



The spinal cord + fast reflexes

CONNECTS

brain ↔ body parts

CARRIES

messages both ways

REFLEX!

A quick, automatic response that helps protect you.



Nerves carry signals at high speed



SENSE → PROCESS → RESPOND

Teacher pause: Which part carries the message first?

Systems working together: running



1 NERVOUS SYSTEM

Plans movement and tells muscles what to do.

2 HEART

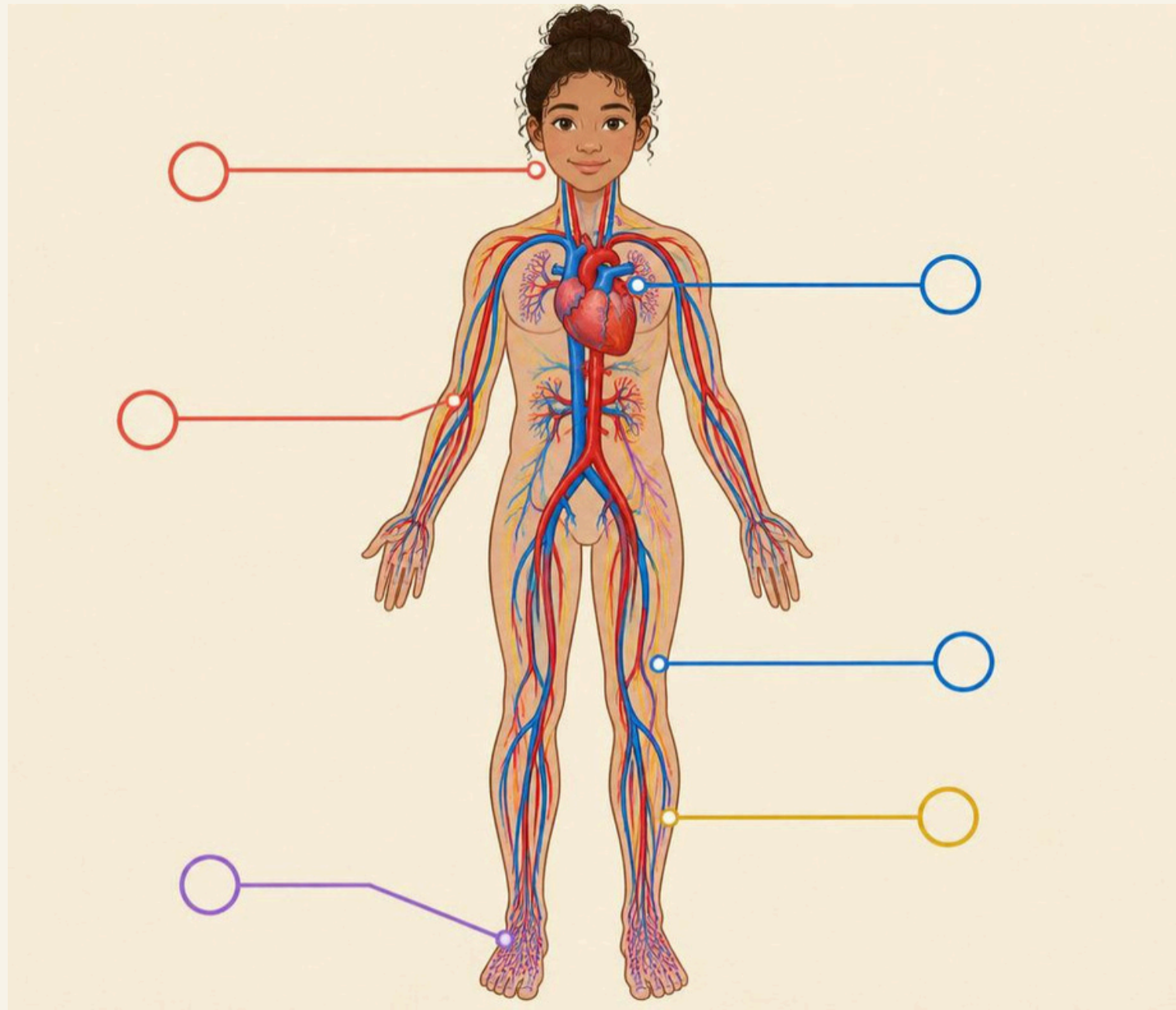
Beats faster to move more blood.

3 CIRCULATORY SYSTEM

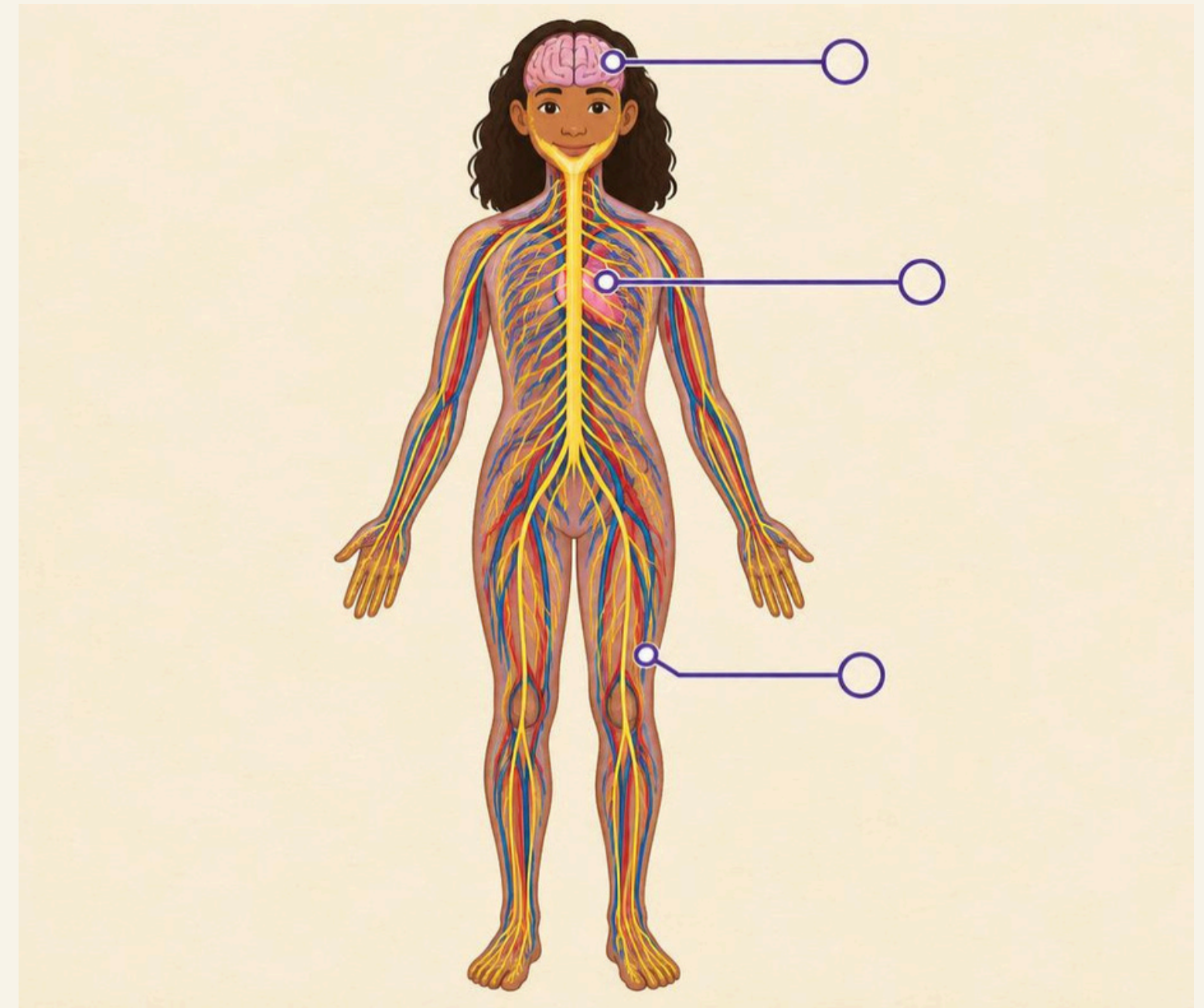
Delivers oxygen + nutrients and carries away waste.

Name that part!

GUESS FIRST • THEN TEACHER
REVEALS



CIRCULATORY



NERVOUS

What does it do?

HEART

- A Pumps blood
- B Controls thoughts
- C Carries nerve messages

BRAIN

- A Pumps blood
- B Controls body activities
- C Carries oxygen

Which system?

Heart pumps blood

☐ CIRCULATORY ☐ NERVOUS ☐ BOTH

Brain processes information

☐ CIRCULATORY ☐ NERVOUS ☐ BOTH

Nerves carry messages

☐ CIRCULATORY ☐ NERVOUS ☐ BOTH

Blood transports oxygen

☐ CIRCULATORY ☐ NERVOUS ☐ BOTH

Running and responding

☐ CIRCULATORY ☐ NERVOUS ☐ BOTH

Point, discuss, then circle your choice.

Assessment and key takeaways

5 QUICK CHECKS

1. What pumps blood?
2. Which vessels go away from the heart?
3. What is the body's control center?
4. What carries messages to body parts?
5. Name one way the systems work together.

- ✓ Circulatory transports materials
- ✓ Heart pumps; vessels carry blood
- ✓ Nervous controls + coordinates
- ✓ Brain is the control center
- ✓ Spinal cord + nerves carry messages
- ✓ Both systems keep you functioning

EXIT REFLECTION

Thank you! Any questions?

Tell a partner: one new thing you learned about how your body works.

